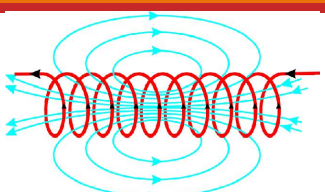
Magnetic Effects of Electric Current

Class X · Physical Science · Chapter 10 (Electromagnetism)

Complete classroom deck: every derivation worked step by step with reasons, labelled diagrams, solved numericals, all textbook questions.

Figures: SCERT Class X Physical Science textbook, pp. 219–248



Chapter roadmap

Part A: Currents make magnetic fields

- 10.1 Oersted experiment
- 10.2 Magnetic field, field lines, flux Φ, flux density B
- 10.3 Field of a straight wire, circular coil, solenoid

Part B: Fields push on currents

- 10.4 $F = qvB\sin\theta$, $F = ILB\sin\theta$, circular motion
- 10.4 Electric motor

Part C: Changing fields make currents

- 10.5 Electromagnetic induction
- 10.5 Faraday's law, Lenz's law
- 10.5 Derivation: motional emf $\varepsilon = Blv$
- 10.5 Applications, magnetic levitation
- 10.5 AC and DC generators

How to use this deck

Badges **1M** **2M** **4M** show usual marks. **Derivation** frames show *step, working, reason*. Answers to **Teacher asks** appear on the next click.

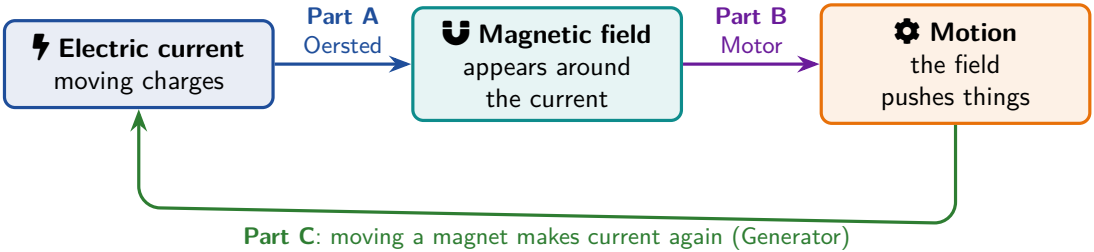
Recall and warm-up

- A magnet has two poles, N and S. Like poles repel, unlike poles attract.
- A freely suspended magnet (compass needle) rests along the **north–south** direction.
- A **field force** acts without contact (gravity, electric, magnetic).
- Electric current = charges in motion: $I = \dfrac{Q}{t}$.
- Electric power $P = \varepsilon I$; work $W = Fs$; power $P = Fv$.

Warm-up questions

- How do electric motors and calling bells work?
- How does an electromagnet work?
- Is there any relation between electricity and magnetism?
- Can we produce electricity from magnetism?

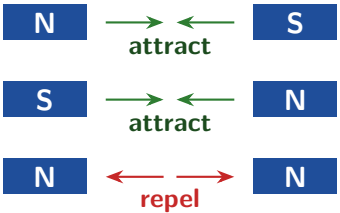
The big picture: one story in three steps



Remember this one line

Electricity and magnetism are **two faces of the same thing**. Current makes magnetism; magnetism pushes current; changing magnetism makes current.

Basics 1: What is a magnet?



- Every magnet has **two poles**: North (N) and South (S).
- **Unlike** poles attract. **Like** poles repel.
- You can never get a single pole: cut a magnet and each piece has both N and S.
- Magnets attract iron, nickel and cobalt.

Think of it like this

A **compass needle** is a tiny magnet. Its N end always swings to the Earth's geographic north because the Earth itself acts like a giant magnet.

Basics 2: What is a “field”?

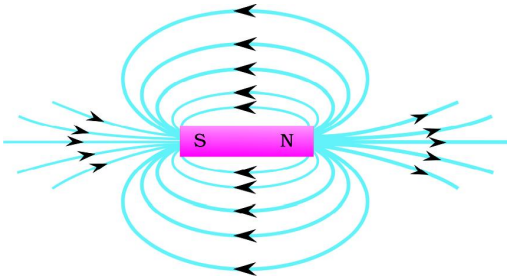
Kitchen smell analogy

Cooking in a kitchen: you smell it **without touching** the food.

- Strong smell near the stove; weak far away $\Rightarrow$ **strength** changes with place.
- Smell drifts *from* the stove *towards* you $\Rightarrow$ it has a **direction**.

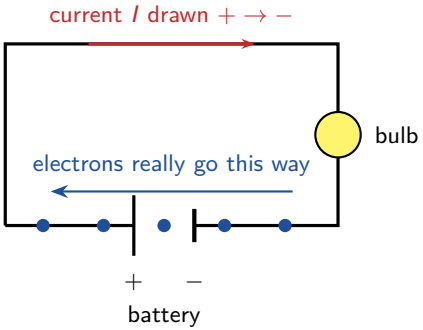
Field

Space around an object where it can push or pull other things **without touching**. A field has **strength** and **direction** at every point.



Magnetic field of a bar magnet. Each curve is a **field line**: the arrow shows the direction; crowded lines mean a stronger field.

Basics 3: What is electric current?



- Current is the **flow of charge**, like water flowing in a pipe.
- Battery = pump, wire = pipe, charge = water.

Current

$$I = \frac{Q}{t} \quad \text{unit: ampere (A)}$$

1 A = 1 coulomb of charge per second.

Direction convention

We **draw** current from + to − (old convention). Electrons really go the other way. Always follow the **drawn** direction in this chapter.

Basics 4: Reading arrows into and out of the page

towards you
you see the **tip**



away from you
you see the **tail feathers**



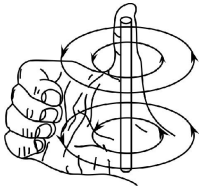
- $\odot$ (dot) $\Rightarrow$ pointing **out of the page**, towards you.
- $\otimes$ (cross) $\Rightarrow$ pointing **into the page**, away from you.
- A page full of $\times$ means a magnetic field going **into** the page; full of $\cdot$ means **out**.

Memory trick

Dot = **D**ooming at you (the *tip* is coming at you).
Cross = **C**arrying away (you see the *tail*, it is leaving).

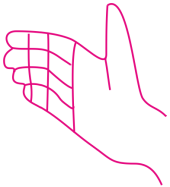
Basics 5: Your right hand is the tool of this chapter

Rule 1: thumb rule



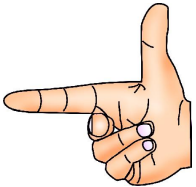
Straight wire.
Thumb = current.
Curled fingers = field.

Rule 2: coil rule



Coil or solenoid.
Curled fingers = current.
Thumb = N pole.

Rule 3: force rule



Force on current/charge.
Forefinger = current I (or v),
middle finger = field B,
thumb = force F.

Always ask: "What do I know, and what do I want?"

Know the **current**, want the **field**? Rule 1 or 2. Know the **current and field**, want the **push**? Rule 3. Use your **right** hand only.

Basics 6: Symbols and units you will meet

Quantity	Symbol	SI unit	In plain words
Electric current	I	ampere (A)	how fast charge flows
Charge	q or Q	coulomb (C)	amount of electric charge
Velocity	v	m/s	speed with direction
Magnetic flux density	B	tesla (T)	strength of the field
Magnetic flux	Φ	weber (Wb)	total field passing through an area
Force	F	newton (N)	a push or pull
Induced emf	ε	volt (V)	electrical "push" that drives current

Unit tip

$1\,\mathrm{T} = 1\,\mathrm{Wb/m^2}$. Convert first: cm $\rightarrow$ m, $\mathrm{cm^2} \rightarrow \mathrm{m^2}$ (divide by 10^4), before substituting into formulas.

Maths 1: Powers of ten (scientific notation)

Why?

Physics numbers can be huge or tiny. Charge of a proton: 0.00000000000000000016 C. Write it as $\mathbf{1.6 \times 10^{-19}}$ C.

- $10^3 = 1000,\ 10^{-3} = \dfrac{1}{1000} = 0.001$
- Move the point **left** $\Rightarrow$ power goes **up**; move **right** $\Rightarrow$ power goes **down**.

Rules

$$10^a \times 10^b = 10^{a+b} \qquad \frac{10^a}{10^b} = 10^{a-b}$$

Worked example

$(2 \times 10^6) \times (1.6 \times 10^{-19}) \times 0.5$

Numbers: $2 \times 1.6 \times 0.5 = 1.6$

Powers: $10^{6-19} = 10^{-13}$

Answer: $\mathbf{1.6 \times 10^{-13}}$

Maths 2: Changing units before you calculate

Change	How	Example
cm $\rightarrow$ m	divide by 100	20 cm $= 0.2$ m
mm $\rightarrow$ m	divide by 1000	5.7 mm $= 5.7 \times 10^{-3}$ m
$\text{cm}^2 \rightarrow \text{m}^2$	divide by 10^4 (not 100!)	4 $\text{cm}^2 = 4 \times 10^{-4}$ m^2
minutes $\rightarrow$ s	multiply by 60	0.2 min $= 12$ s

Why 10^4 for area?

1 cm $= \frac{1}{100}$ m, so 1 $\text{cm}^2 = \frac{1}{100} \times \frac{1}{100} = \frac{1}{10\,000}$ m^2. Area squares the conversion factor.

Maths 3: Rearranging a formula (the balance method)

Golden rule

An equation is a balance. Whatever you do to one side, do the **same** to the other.

Start with $F = ILB$.

- Find B: divide both sides by $IL \Rightarrow B = \dfrac{F}{IL}$
- Find I: divide both sides by $LB \Rightarrow I = \dfrac{F}{LB}$
- Find L: divide both sides by $IB \Rightarrow L = \dfrac{F}{IB}$

Example from the chapter

$r = \dfrac{mv}{Bq}$. Find v.

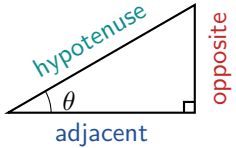
Multiply both sides by Bq: $rBq = mv$

Divide both sides by m: $v = \dfrac{rBq}{m}$

Steps for every numerical

1. Write the formula. 2. Convert units. 3. Rearrange if needed. 4. Substitute. 5. Calculate and write the **unit**.

Maths 4: Sine and cosine of an angle



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

θ	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0

Where we use them

$F = qvB\sin\theta$: θ is the angle between v and B. At 90° the force is maximum; at 0° it is zero.

$\Phi = BA\cos\theta$: θ is the angle between B and the **normal** to the plane. At 0° the flux is maximum.

Maths 5: Area, circles and time period

Shapes

Rectangle: $A = \text{length} \times \text{breadth}$
Circle: $A = \pi r^2$, circumference $= 2\pi r$ ($\pi \approx 3.14$)

Example

Loop of radius 0.1 m: $A = 3.14 \times 0.1^2 = 0.0314$ m^2.

Speed, distance, time

$$v = \frac{\text{distance}}{\text{time}} \Rightarrow \text{time} = \frac{\text{distance}}{v}$$

Time period T

Time for one full circle: $T = \dfrac{2\pi r}{v}$. We use this in the circular-path derivation.

Maths 6: "Change" and "rate of change" (Δ)

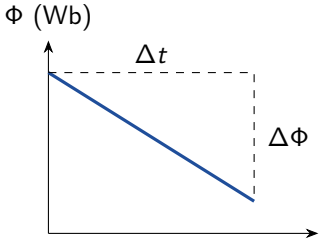
- Δ (delta) means **change** = final − initial.
- $\Delta\Phi = \Phi_{\text{final}} - \Phi_{\text{initial}}$, $\quad \Delta t = t_{\text{final}} - t_{\text{initial}}$
- **Rate of change** = change ÷ time taken, e.g. $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$.

Example

Flux falls from 0.5 Wb to 0.1 Wb in 0.2 s.

$\Delta\Phi = 0.1 - 0.5 = -0.4$ Wb

$|\varepsilon| = \dfrac{0.4}{0.2} = \mathbf{2}$ V



Slope of a flux–time graph $= \dfrac{\Delta\Phi}{\Delta t}$ = induced emf. A steeper line means a bigger emf; a flat line means zero emf.

Maths 7: Proportion, "if this doubles, what happens?"

Formula	If this doubles...	That
$F = ILB$	B doubles	F doubles
$F = ILB$	I and L both double	F becomes $4\times$
$r = \dfrac{mv}{Bq}$	B doubles	r **halves**
$\varepsilon = Blv$	v doubles	ε doubles
$\Phi = BA\cos\theta$	A doubles	Φ doubles

Two kinds of relation

Direct ($F \propto B$): both grow together. **Inverse** ($r \propto \frac{1}{B}$): one grows, the other shrinks.

Exam Tip

Spot the variable on top (direct) or underneath (inverse) in the formula.

Maths check: try these

Quick maths drill

1. Write 0.0005 in powers of ten.
2. Convert 50 cm^2 to m^2.
3. $F = ILB$: find L if $F = 6$ N, $I = 3$ A, $B = 0.5$ T.
4. Value of $\sin 30^\circ$ and $\cos 60^\circ$?
5. Flux changes by 0.6 Wb in 0.3 s. Find ε.

Answers

1. 5×10^{-4}
2. $5 \times 10^{-3}\ \text{m}^2$
3. $L = \dfrac{6}{3 \times 0.5} = 4$ m
4. both $\frac{1}{2}$
5. $\varepsilon = \dfrac{0.6}{0.3} = 2$ V

Quick check before we start

Try these (answers on the next click)

1. Two N poles are brought close. Attract or repel?
2. Charge 6 C flows in 2 s. Find the current.
3. An arrow shows $\otimes$. Is it going into or out of the page?
4. Which hand do we use for the rules?

Answers

1. Repel (like poles).
2. $I = Q/t = 6/2 = 3$ A.
3. Into the page.
4. The right hand.

Part A: Currents make magnetic fields

By the end of this part you can…

- Describe what Oersted observed and what it proves.
- Draw and read **magnetic field lines**; tell a strong field from a weak one.
- Explain **flux** Φ and **flux density** B, and use $\Phi = BA\cos\theta$.
- Find the field direction around a wire, a coil and a solenoid with the **right hand rule**.

10.1 Oersted experiment (Activity 1)

2M 4M

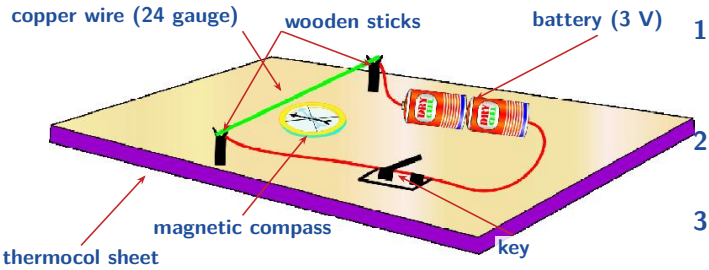


fig-1: current-carrying wire above a compass

1. Wire stretched through slits of two sticks, in series with battery and key. Compass below the wire.
2. Bring a bar magnet near: needle deflects (magnetic force).
3. Remove the magnet, switch **on**: needle **deflects again**!
4. Switch off: needle returns to N–S.

Conclusion

A **current-carrying wire produces a magnetic field** around it. Electricity and magnetism are related.

Do you know? Hans Christian Oersted

1M



H. C. Oersted (1777–1851)

- In a lecture in April 1820 he placed a compass needle under a wire and switched on the current: the needle **deflected**.
- Showed that electricity and magnetism are **interconnected**: the start of **electromagnetism**.
- Led to the dynamo, the electric motor, radio, television and fibre optics.
- The unit of magnetic field strength is named **oersted** in his honour.

10.2 Magnetic field of a bar magnet (Activity 2)

1M 2M

Activity 2: compass around a bar magnet

Place a compass at many points around a bar magnet on paper. Mark the direction of the needle at each point (S → N of the needle).

- The needle points in **different directions** at different places ⇒ the field has **direction** that varies from point to point.
- Far from the magnet the needle points almost N–S ⇒ **strength** decreases with distance.
- Above the magnet the needle is also deflected ⇒ the field is **three-dimensional**.

Field

A region where a force acts on an object **without physical contact**.

Magnetic field

The region surrounding a magnet in which its magnetic force can be felt. It is characterised by **strength** and **direction**.

10.2.1 Magnetic field lines (Activity 3)

1M 2M Diagram

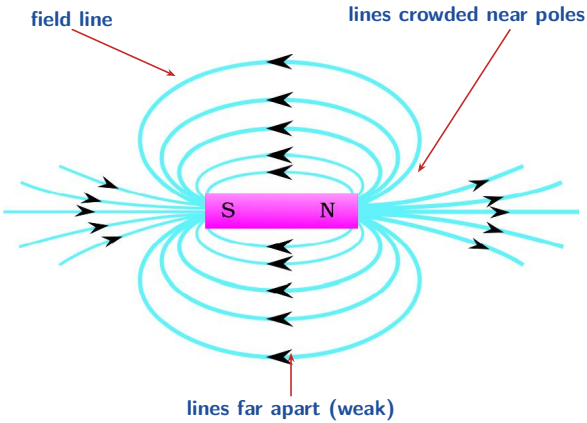


fig-2: field lines of a bar magnet (outside: N → S)

Plotting: put the compass at N, mark a dot at the tip of the needle, move the compass to the dot, repeat until you reach S. Join the dots.

- Field lines are **imaginary** curves.
- **Tangent** at any point gives the direction of the field.
- **Crowded** lines ⇒ strong field; **spread** lines ⇒ weak field.
- **Non-uniform field**: strength **or** direction changes.
  Uniform field: strength **and** direction constant (parallel, equally spaced lines).

10.2.2 Magnetic flux and flux density

1M 2M

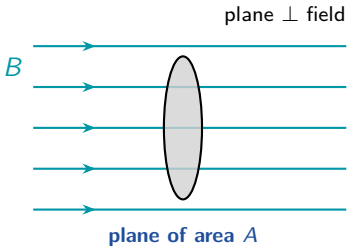


fig-3(a): lines crossing a plane placed perpendicular to a uniform field

Magnetic flux Φ

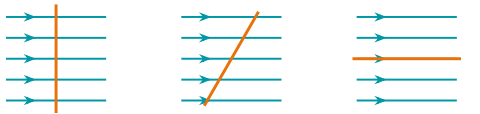
The number of field lines passing through a plane of area A held **perpendicular** to the field. SI unit: **weber (Wb)**.

Magnetic flux density B

Flux passing through **unit area** held perpendicular to the field. Also called **magnetic induction**.

$$B = \frac{\Phi}{A} \qquad \text{unit: } \mathrm{Wb/m^2} = \text{tesla (T)}$$

Flux in simple words: rain through a window



face-on: **most** lines tilted: **fewer** edge-on: **none**

Orange line = the window seen edge-on; blue lines = field lines (rain).

Analogy

Rain = field lines. Window = area A.

- Hold the window **face-on**: the most rain passes.
- **Tilt** it: less passes.
- **Edge-on**: no rain passes.

Flux Φ = how much rain passes. **Flux density** B = how heavy the rain is.

Derivation 1: flux through a tilted plane, $\Phi = BA\cos\theta$

2M

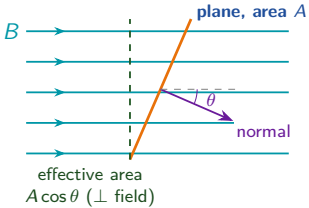


fig-3(b): edge view of the plane

1. $B = \dfrac{\Phi}{A_\perp}$ — B is always defined with the area **perpendicular** to the field
2. $A_\perp = A\cos\theta$ — θ = angle between B and the normal; only this part of the plane faces the field
3. $B = \dfrac{\Phi}{A\cos\theta}$ — substitute step 2 in step 1
4. $\Phi = BA\cos\theta$ — multiply both sides by $A\cos\theta$

Special cases

$\theta = 0^\circ$ (plane $\perp$ field): $\Phi = BA$ (maximum)
$\theta = 90^\circ$ (plane $\parallel$ field): $\Phi = BA\cos 90^\circ = 0$

Solved numericals: flux and flux density

2M

Q1. $B = 0.5$ T through a loop of area 0.02 m^2, loop $\perp$ field. Find Φ.

1. Plane $\perp$ field $\Rightarrow \theta = 0^\circ$, $\cos 0^\circ = 1$.
2. $\Phi = BA = 0.5 \times 0.02 = \boxed{0.01 \text{ Wb}}$

Q2. Same loop turned so the normal makes 60° with B.

1. $\Phi = BA\cos 60^\circ = 0.01 \times \frac{1}{2} = \boxed{0.005 \text{ Wb}}$

Q3. Flux of 2×10^{-3} Wb crosses 4 cm^2 normally. Find B.

1. $A = 4 \text{ cm}^2 = 4 \times 10^{-4} \text{ m}^2$ ($1 \text{ cm}^2 = 10^{-4} \text{ m}^2$)
2. $B = \dfrac{\Phi}{A} = \dfrac{2 \times 10^{-3}}{4 \times 10^{-4}} = \boxed{5 \text{ T}}$

Textbook question

What is the flux through a plane held **parallel** to the field?

Zero: no field line crosses it ($\theta = 90^\circ$).

10.3.1 Field of a straight current-carrying wire (Activity 4) 2M 4M

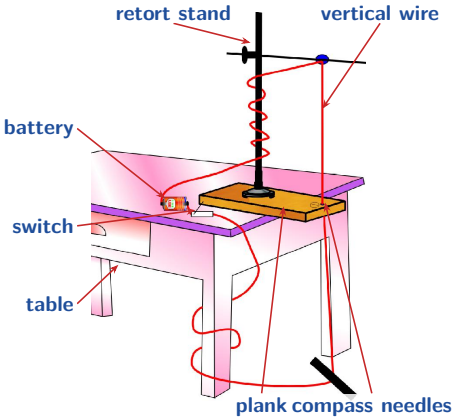


fig-4(a)

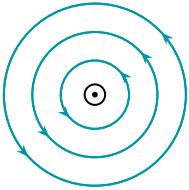
1. Pass a vertical wire through a hole in a plank; place 6–10 compass needles on a circle around the hole.
2. Switch on (3 V battery).
3. The needles set along **tangents to the circle**.
4. Sprinkle iron filings: they form **concentric circles**.

Conclusion

Field lines of a straight wire are **concentric circles** centred on the wire, in a plane perpendicular to it. They are **closed loops**. The field is stronger near the wire (circles closer).

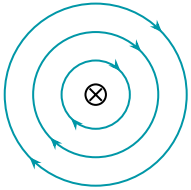
Direction of the field: right hand thumb rule

1M 2M



current **out of** page
field **anticlockwise**

fig-4(b)



current **into** page
field **clockwise**

fig-4(c)

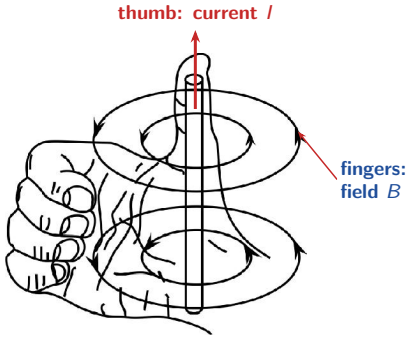


fig-4(d)

Right hand thumb rule

Grasp the wire with the right hand with the **thumb along the current**; the **curled fingers** give the direction of the field lines.

Symbols: ⊙ current/field out of the page (arrow tip); ⊗ into the page (arrow tail).

10.3.2 Field of a circular coil (Activity 5)

2M 4M

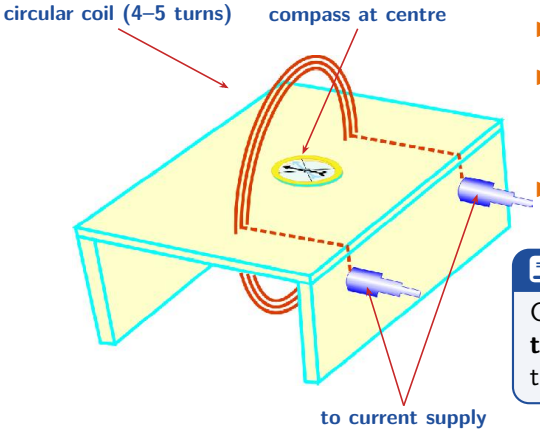


fig-5(a)

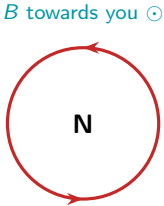
- Plot field lines with a compass around the coil.
- At the centre the needle points **perpendicular to the plane of the coil**: the field there is along the axis.
- Bring a compass near one face: its S pole points to the face ⇒ that face acts as a **N pole**.

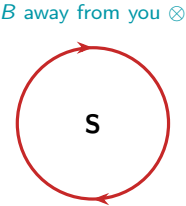
Right hand rule for a coil

Curl the right-hand fingers along the **current**; the **thumb** gives the direction of the **field** (points out of the N face).

Coil: which face is North?

1M 2M





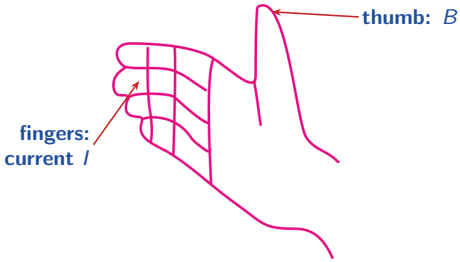


fig-5(b): right hand rule for a coil

Exam Tip

Memory aid: letters **N** and **S** drawn with arrows at their ends run anti-clockwise and clockwise respectively.

10.3.3 Field of a solenoid (Activity 6)

2M 4M

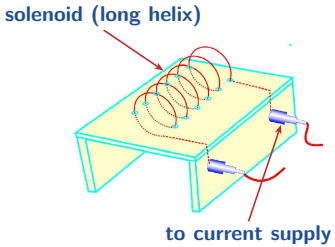


fig-6(a)

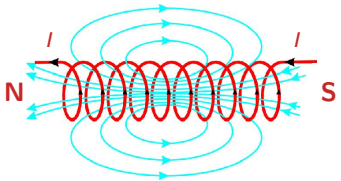


fig-6(b): field lines (iron filings)

Solenoid

A long wire wound in a **close-packed helix**.

- Iron filings show a pattern like that of a **bar magnet**: a solenoid behaves like a bar magnet.
- One end is N, the other S (find with the right hand rule).
- Outside: field lines go **N** $\rightarrow$ **S**; inside: **S** $\rightarrow$ **N**, strong and nearly uniform (parallel lines).
- Lines outside are continuous with those inside $\Rightarrow$ field lines are **closed loops**. Same for a bar magnet.

Big idea

Moving charges (currents) produce magnetic fields.

Part B: Fields push on currents

☑ By the end of this part you can…

- Find the **direction and size** of the force on a moving charge, $F = qvB\sin\theta$.
- Show why a charge in a field moves in a **circle**: $r = mv/Bq$.
- Derive $F = ILB\sin\theta$ for a current-carrying wire.
- Explain how an **electric motor** keeps turning.

10.4 Force on a moving charge (Activity 7)

1M 2M

Activity 7: magnet near an old CRT TV

The picture gets **distorted**; move the magnet away and it clears.

The picture is made by a beam of **moving electrons**. The magnetic field exerts a force on them and changes their path: the **magnetic force**.

Magnetic force on a charge

$$F = qvB \quad (\vec{v} \perp \vec{B}) \qquad F = qvB\sin\theta$$

θ = angle between $\vec{v}$ and $\vec{B}$.

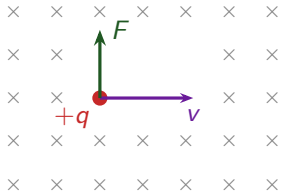


fig-7: charge moving $\perp$ to B

Special cases of $F = qvB\sin\theta$

$\theta = 90°$: $F = qvB$ (maximum)
$\theta = 0°$ or $180°$ (moving along or against B): $\sin\theta = 0 \Rightarrow F = 0$
$v = 0$ (charge at rest): $F = 0$

Direction of the magnetic force: right hand rules

1M 2M

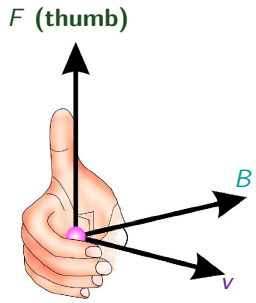


fig-8(a): fingers along v, curl towards B; thumb gives F

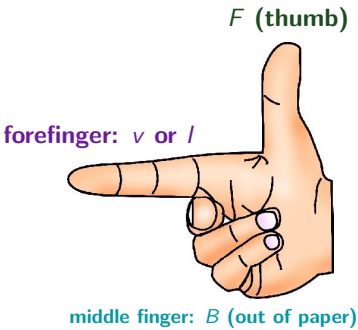


fig-8(b): three fingers mutually perpendicular

Exam Tip

F is always **perpendicular to both** v and B. Rules are for a **positive** charge; for a **negative** charge (electron) find the direction for $+q$, then **reverse** it.

Derivation 2: circular path of a charge, $r = \dfrac{mv}{Bq}$, $T = \dfrac{2\pi m}{Bq}$

4M

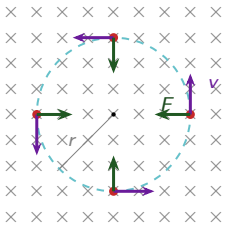


fig-E-1: $F \perp v$ always, towards the centre

1. $F = qvB$ — $v \perp B$, so $\sin 90^\circ = 1$
2. $F \perp v$ — force never changes the speed, only the direction $\Rightarrow$ uniform circular motion
3. $\dfrac{mv^2}{r} = qvB$ — magnetic force supplies the centripetal force
4. $r = \dfrac{mv}{Bq}$ — multiply by r, cancel one v, divide by qB
5. $T = \dfrac{2\pi r}{v}$ — time = circumference ÷ speed
6. $T = \dfrac{2\pi}{v} \cdot \dfrac{mv}{Bq} = \dfrac{2\pi m}{Bq}$ — substitute r; v cancels

Note: T does **not** depend on speed: a faster particle moves in a bigger circle in the same time.

Solved numericals: force on a moving charge

2M

Q1. A proton (1.6×10^{-19} C) moves at 2×10^{6} m/s $\perp$ to $B = 0.5$ T. Find F.

1. $F = qvB \quad (\theta = 90^\circ)$
2. $F = 1.6 \times 10^{-19} \times 2 \times 10^{6} \times 0.5$
3. $F = \boxed{1.6 \times 10^{-13}\ \text{N}}$

Q2. Same proton moves at 30° to B.

1. $F = qvB\sin 30^\circ = 1.6 \times 10^{-13} \times \frac{1}{2}$
2. $F = \boxed{8 \times 10^{-14}\ \text{N}}$

Q3. Electron ($m = 9.1\times10^{-31}$ kg) at 10^{6} m/s $\perp$ to $B = 10^{-3}$ T. Radius?

1. $r = \dfrac{mv}{Bq} = \dfrac{9.1 \times 10^{-31} \times 10^{6}}{10^{-3} \times 1.6 \times 10^{-19}}$
2. $= \dfrac{9.1 \times 10^{-25}}{1.6 \times 10^{-22}} = \boxed{5.7 \times 10^{-3}\ \text{m}} \approx 5.7\ \text{mm}$

Teacher asks

A charge moves parallel to B. Force?
Zero, since $\sin 0^\circ = 0$.

Derivation 3: force on a current-carrying wire, $F = ILB$

4M

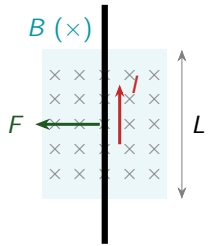


fig-9: only length L lies in the field

Exam Tip

$\parallel B$: $F = 0$. $\perp B$: $F = ILB$ (max). In fig-9 the force on the upward current is to the **left** (right hand rule).

1. $F_0 = qvB$ — force on one charge moving with drift velocity $v \perp B$
2. $F = QvB \cdots (1)$ — add up for the total charge Q inside the field
3. $t = \frac{L}{v} \Rightarrow v = \frac{L}{t} \cdots (2)$ — time for the charge to cross the length L
4. $F = \frac{QLB}{t} = \frac{Q}{t}LB \cdots (3)$ — put (2) in (1)
5. $I = \frac{Q}{t}$ — definition of current
6. $F = ILB \cdots (4)$ — put step 5 in (3)
7. $F = ILB\sin\theta \cdots (5)$ — θ = angle between current and field

Activity 8: wire in a horseshoe magnet

4M: experiment

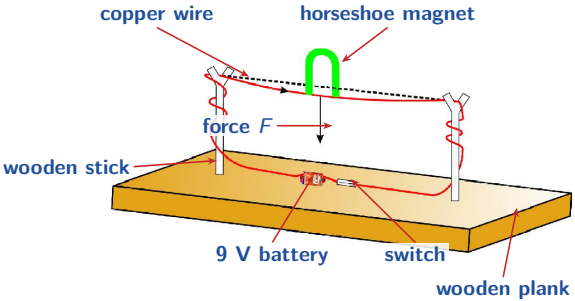


fig-10

1. Stretch a copper wire between two split sticks; connect to a 9 V battery and switch.
2. Switch on and bring a horseshoe magnet around the wire.
3. The wire is **deflected**; the direction agrees with the right hand rule.
4. Reverse the magnet's poles, or reverse the current: the deflection **reverses**.

Conclusion

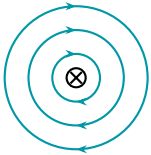
A current-carrying wire in a magnetic field experiences a force; its direction depends on the directions of the current and of the field.

Why does the wire move? Combining the fields

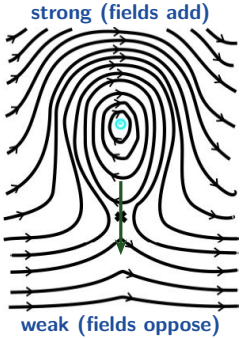
2M 4M



(a) magnet's field



(b) wire, current into page (clockwise)



(c) resultant: wire pushed to weak side

- Above the wire, its clockwise field points **with** the magnet's field: strong.
- Below, it points **against**: weak.
- The non-uniform field pushes the wire from the strong to the **weak** region.

Check with the rule: I into page, B left to right $\Rightarrow$ F downward. fig-11(a),(b),(c)

Solved numericals: force on a wire

2M

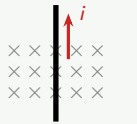
Textbook App. Q4. $F = 8$ **N on a 20 cm conductor** $\perp$ **to** B, $I = 40$ **A. Find** B.

1. $L = 20$ cm $= 0.2$ m
2. $F = ILB \Rightarrow B = \dfrac{F}{IL}$
3. $B = \dfrac{8}{40 \times 0.2} = \dfrac{8}{8} = \boxed{1\ \text{T}}$

Q2. $I = 5$ **A,** $L = 0.5$ **m,** $B = 0.2$ **T, wire at** 30° **to** B.

1. $F = ILB \sin 30^\circ = 5 \times 0.5 \times 0.2 \times \frac{1}{2} = \boxed{0.25\ \text{N}}$

Textbook App. Q3 (fig-Q5). Wire along its length, current i **upward,** B **into page, length** L **in field.**



1. $\theta = 90^\circ \Rightarrow F = iLB$
2. Right hand rule: forefinger up (i), middle finger into page (B) $\Rightarrow$ thumb points **to the left**.

Teacher asks

Force on a wire kept **parallel** to B?
Zero: every charge moves along the field.

Motor in simple words

Three ideas, in order

1. A wire carrying current inside a magnetic field gets **pushed** sideways.
2. A coil has two long sides with currents in **opposite** directions, so one side is pushed **up** and the other **down**: the coil **spins**.
3. After half a turn the pushes would stop the spin. The **split ring (commutator)** reverses the current at that moment, so the coil keeps turning the same way.

Pedal analogy

Cycling: push down with one foot, then the other. The commutator is you switching feet at exactly the right time.

10.4.1 Electric motor: forces on a coil

4M

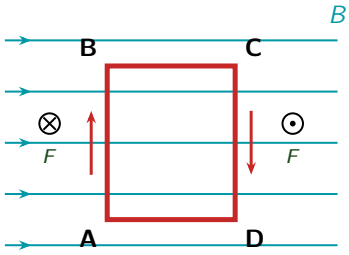


fig-12(a): current A→B→C→D; F on AB into page, on CD out of page

1. AB and CD are always **perpendicular** to $B \Rightarrow$ force ILB on each.
2. Right hand rule: on AB (current up) force **into** the page; on CD (current down) force **out of** the page.
3. BC and DA: force zero when parallel to B; otherwise equal and opposite (up on BC, down on DA).
4. Forces on AB and CD: equal, opposite (equal currents in opposite directions) $\Rightarrow$ **net force = 0**.
5. But they act along **different lines** $\Rightarrow$ they form a **couple** and **rotate** the coil, like two hands opening a bottle cap.

Electric motor: continuous rotation

4M Diagram

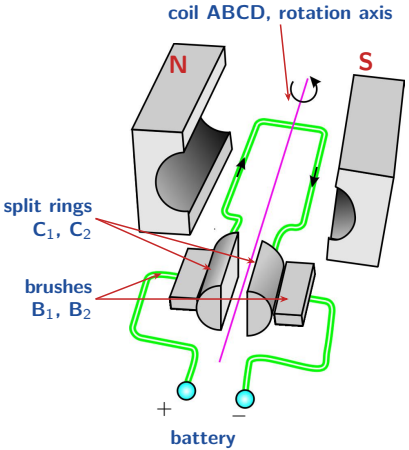


fig-12(c): simple electric motor

- Without reversal: the coil turns to the vertical, overshoots by inertia, then the forces turn it **back**; it rocks to and fro.
- **Fix:** reverse the current in the coil **every half rotation**.
- The coil ends go to rings C_1, C_2 that rotate with it; fixed brushes B_1, B_2 connect to the battery.
- After each half turn the rings swap brushes ⇒ current in the coil reverses ⇒ rotation continues in the **same direction**.

Energy

Motor: **electrical energy** → **mechanical energy**.

Do you know? Fleming's left hand rule

Stretch left-hand thumb, forefinger and middle finger mutually perpendicular: **forefinger** = field, **middle finger** = current, **thumb** = force (motion).

Part C: Changing fields make currents

☑ By the end of this part you can…

- Explain **electromagnetic induction** with the magnet-and-coil activity.
- State **Faraday's law** and **Lenz's law** and apply them.
- Derive the motional emf $\varepsilon = Blv$.
- Explain how **AC and DC generators** and magnetic levitation work.

10.5 Electromagnetic induction (Activity 9)

2M 4M

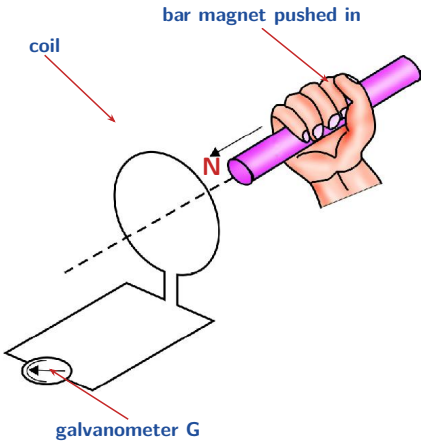


fig-13

What we do	Galvanometer
Magnet at rest near coil	no deflection
N pole moved **towards** coil	deflects one way
N pole moved **away**	deflects the **other** way
S pole used instead	deflections reversed
Coil moved, magnet fixed	also deflects

Electromagnetic induction

Producing a current in a coil by **relative motion** between the coil and a magnet (changing flux). The current is the **induced current**; it is driven by an **induced emf**.

Induction in simple words: only *change* counts

Key idea

A magnet sitting still inside a coil makes **no** current, however strong it is. A magnet that is **moving** (or a field that is changing) makes a current.

- **Faster** movement $\Rightarrow$ **larger** induced emf.
- **More turns** in the coil $\Rightarrow$ larger induced emf.
- **Stronger** magnet $\Rightarrow$ larger induced emf.
- Reverse the motion $\Rightarrow$ the current flows the **opposite** way.

10.5.1 Faraday's law

1M 2M 4M

Faraday's law (first form)

Whenever there is a **continuous change of magnetic flux** linked with a closed coil, a current is generated in the coil.

Faraday's law of induction

The induced emf in a closed loop **equals the rate of change of magnetic flux** through it.

Faster change of flux ⇒ greater induced emf.

Formula

$$\varepsilon = \frac{\Delta\Phi}{\Delta t} \quad \cdots (6)$$

Coil of N turns, flux Φ_0 per turn:

$$\Phi = N\Phi_0 \quad \cdots (7) \qquad \varepsilon = N\frac{\Delta\Phi_0}{\Delta t}$$

Exam Tip

No change of flux ⇒ no emf, however strong the field.

Lenz's law in simple words: nature resists change

Think of a heavy door

Push a heavy door and you **feel it push back**. The coil does the same: when you push a magnet in, the induced current makes the coil push **back**.

- Induced current **always opposes** what causes it.
- Magnet approaching $\Rightarrow$ coil face becomes the **same pole** (repels).
- Magnet leaving $\Rightarrow$ coil face becomes the **opposite** pole (attracts).

Why must it oppose?

If it helped the motion, the magnet would speed up by itself and give free energy forever. That breaks **conservation of energy**.

10.5.2 Lenz's law: the direction of the induced current

2M 4M

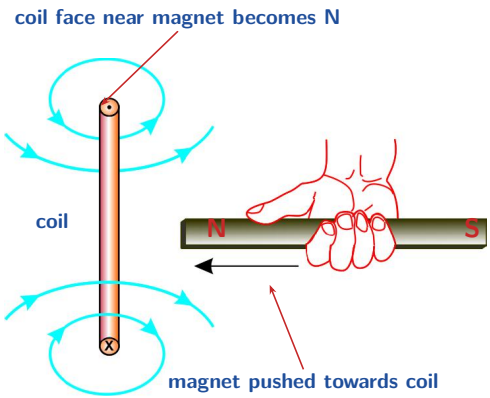


fig-14: N faces N ⇒ the coil repels the magnet

Reasoning (conservation of energy):

1. Suppose the induced current made the coil's near face **S**. It would **attract** the approaching N pole; the magnet would speed up and gain kinetic energy **from nothing**, and electrical energy would also appear. Impossible.
2. So the near face must be **N**: it **repels** the magnet. We must do work to push it; that work becomes electrical energy.
3. Pulling the magnet away: the near face becomes **S** and attracts, again opposing the motion.

Lenz's law

The induced current flows in a direction that **opposes the change in flux** that produces it.

10.5.3 Derivation 4, part 1: the set-up

4M

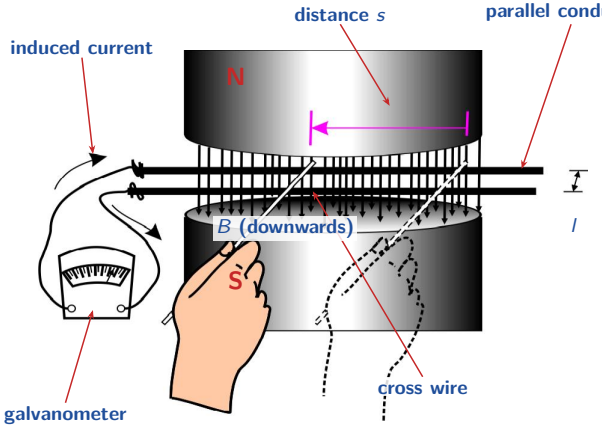
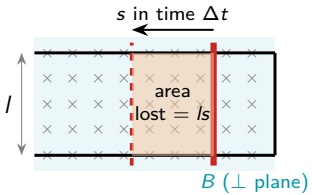


fig-15

- Two parallel bare conductors, l metres apart, in a uniform field B (perpendicular to their plane), joined through a galvanometer.
- A bare **cross wire** rests across them, closing the loop.
- Move the cross wire to the **left**: needle deflects one way. To the **right**: the other way.
- It moves a distance s in time Δt, with speed $v = \dfrac{s}{\Delta t}$.
- A current I flows $\Rightarrow$ there is an emf ε. By **conservation of energy**, the electrical energy comes from the work we do in moving the wire (friction ignored).

Derivation 4, part 2: work done equals flux change × current

4M



top view of the loop

1. $F = BIl \quad \cdots(8)$ — field's force on current I in length l (eq. 4); opposes our push
2. $W = Fs = BIls \quad \cdots(9)$ — our work against it over distance s
3. $\Delta\Phi = B(ls) \quad \cdots(10)$ — loop area falls by ls; $B \perp$ area so $\Phi = BA$
4. $W = (\Delta\Phi)\, I$ — compare (9) and (10): $Bls = \Delta\Phi$
5. $\dfrac{W}{\Delta t} = \dfrac{I\,\Delta\Phi}{\Delta t} \quad \cdots(11)$ — divide by Δt (work per second)
6. $P = \varepsilon I \quad \cdots(12)$ — electric power = emf × current
7. $\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$ — compare (11), (12): **Faraday's law**

Derivation 4, part 3: motional emf, $\varepsilon = Blv$

4M

1. $\frac{W}{\Delta t} = \frac{Fs}{\Delta t} = \frac{Blls}{\Delta t} \quad \cdots (13)$ — divide (9) by Δt
2. $\frac{s}{\Delta t} = v$ — speed of the cross wire
3. $P = Fv = Blv \quad \cdots (14)$ — power = force × velocity
4. $\varepsilon I = Blv$ — (12) and (14) are the same power
5. $\varepsilon = Blv$ — cancel I on both sides

Motional emf

$$\boxed{\varepsilon = Blv}$$

emf between the ends of a conductor of length l moving with speed v perpendicular to a uniform field B.

Note

$\varepsilon = Blv$ is **not** Faraday's law itself: it refers to a **single moving conductor**, not a loop. It is useful whenever a conductor moves in a uniform field.

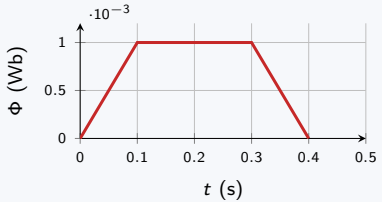
Chain of equations

(8) $F = BIl \rightarrow$ (9) $W = Blls$
(10) $\Delta\Phi = Bls \rightarrow W = I\Delta\Phi$
(11) $\div \Delta t$, (12) $P = \varepsilon I \rightarrow \varepsilon = \frac{\Delta\Phi}{\Delta t}$
(13), (14) $P = Fv = Blv \rightarrow \varepsilon = Blv$

Solved examples: induced emf

2M 4M

Example 1 (textbook). Coil of 400 turns; flux per turn varies as in the graph.



1. 0 to 0.1 s: $\Delta\Phi_0 = 0.001$ Wb, $\Delta t = 0.1$ s
2. $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t} = \dfrac{400 \times 0.001}{0.1} = \boxed{4\text{ V}}$ (maximum)
3. 0.1 s to 0.3 s: flux constant $\Rightarrow \Delta\Phi = 0 \Rightarrow \varepsilon = \boxed{0}$

Example 2 (textbook). $v = 10$ **m/s** $\perp B =$ 0.8 **T,** $\varepsilon = 8$ **V. Find** l**.**

1. $\varepsilon = Blv \Rightarrow l = \dfrac{\varepsilon}{Bv}$
2. $l = \dfrac{8}{0.8 \times 10} = \boxed{1\text{ m}}$

Q3. Flux through a 50-turn coil falls from 0.02 **Wb to 0 in** 0.1 **s (per turn).**

1. $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t} = \dfrac{50 \times 0.02}{0.1} = \boxed{10\text{ V}}$

10.5.4 Applications of electromagnetic induction

2M Textbook App. Q6

Security check (metal detector)

People walk through a large coil carrying a weak AC field. Iron on the body changes the flux linked with the coil; the induced current **triggers an alarm**.

Tape recorder

Tape coated with iron oxide is magnetised unevenly. Moving past a small coil (the head), its changing field induces a current in the coil.

Induction stove

A coil below the cooking surface carries AC $\Rightarrow$ alternating field. It induces an emf and current in the **metal pan**; the pan's resistance turns the current into **heat**, which heats the water.

ATM card swipe

The magnetic strip moves past a coil in the reader; the changing flux induces a current pattern that is read as the card's data.

10.5.5 Magnetic levitation

2M 4M

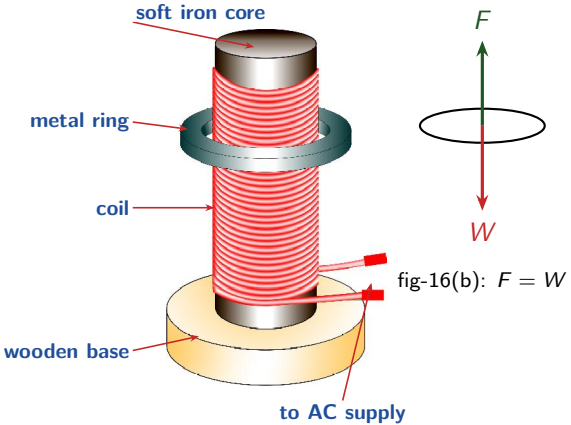


fig-16(a)

fig-16(b): $F = W$

1. AC through the coil: the ring **floats**. Switch off: the ring jumps up.
2. Ring at rest $\Rightarrow$ net force 0 (Newton's 2nd law): an upward force F balances the weight W.
3. AC keeps reversing $\Rightarrow$ the coil's poles reverse $\Rightarrow$ the flux through the ring keeps changing $\Rightarrow$ induced current in the ring.
4. By Lenz's law the ring's face is **like** the coil's top face (coil top S, ring bottom S) $\Rightarrow$ **repulsion** lifts the ring.
5. With **DC** the ring jumps only at switch on/off (flux changes only then) and falls back.

Levitation: why is the ring repelled?

2M

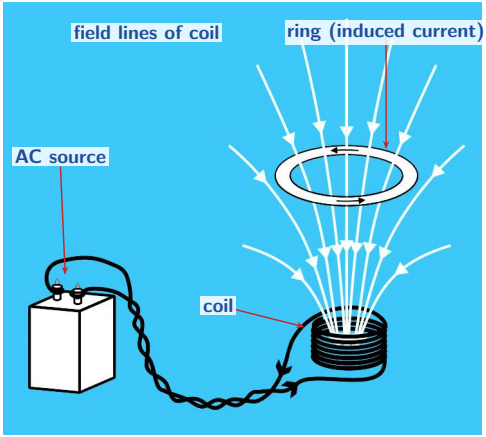


fig-16(c)

1. AC in coil
2. flux through ring changes
3. induced current in ring
4. ring opposes the change
5. repulsion $= W$

current changes size and direction continuously

ring area fixed, field through it changes

Faraday's law

Lenz's law: ring's lower face is like the coil's upper face

upward force balances weight: ring floats

Generator in simple words: a motor run backwards

Motor

Electrical energy $\longrightarrow$ motion
current pushes the coil

Generator

Motion $\longrightarrow$ electrical energy
we spin the coil, current appears

Why it works

Spinning the coil in a magnetic field keeps changing the flux through it, so an emf is induced (Faraday's law). Turn faster $\Rightarrow$ more emf.

4M Diagram

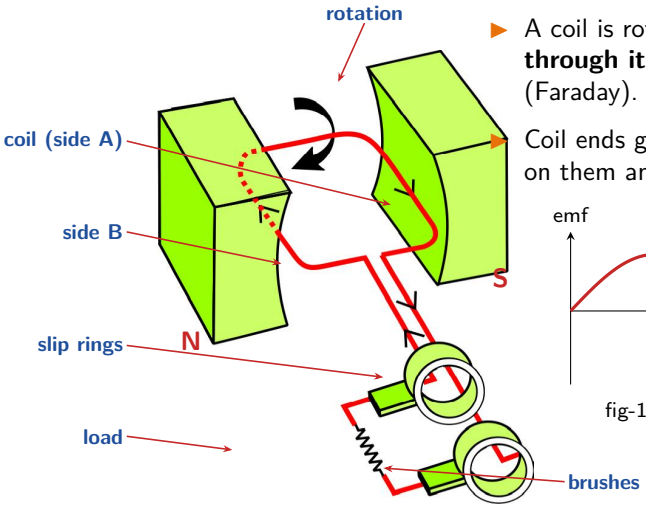


fig-17(a)

- A coil is rotated between the poles of a magnet; the **flux through it changes** continuously ⇒ induced emf (Faraday).

Coil ends go to two **full slip rings**; carbon brushes press on them and take current to the load.

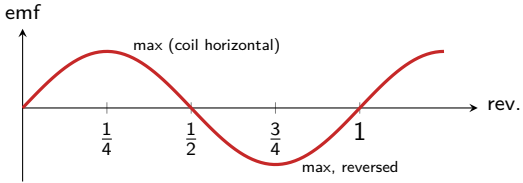


fig-17(b): alternating emf over one rotation

Why does the emf vary? AC and the DC generator

2M 4M

One rotation, step by step:

1. Coil **vertical** (plane $\perp$ field): flux maximum but changing slowest $\Rightarrow$ **emf zero**.
2. First quarter: emf rises; coil **horizontal** (plane $\parallel$ field): flux changing fastest $\Rightarrow$ **emf maximum**.
3. Second quarter: emf falls to zero (vertical again, B on top).
4. Second half: same pattern but sides A and B have swapped places $\Rightarrow$ **emf reversed**.

Alternating current (AC)

Current whose direction reverses periodically; it has a **frequency**.

Generator: **mechanical $\rightarrow$ electrical energy**.

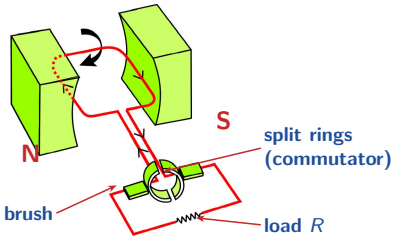


fig-17(c): DC generator

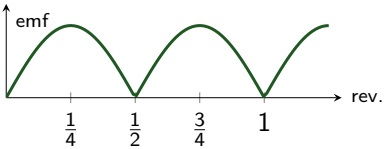


fig-17(d): with half slip rings the current in the load keeps one direction (DC)

Motor vs generator

2M

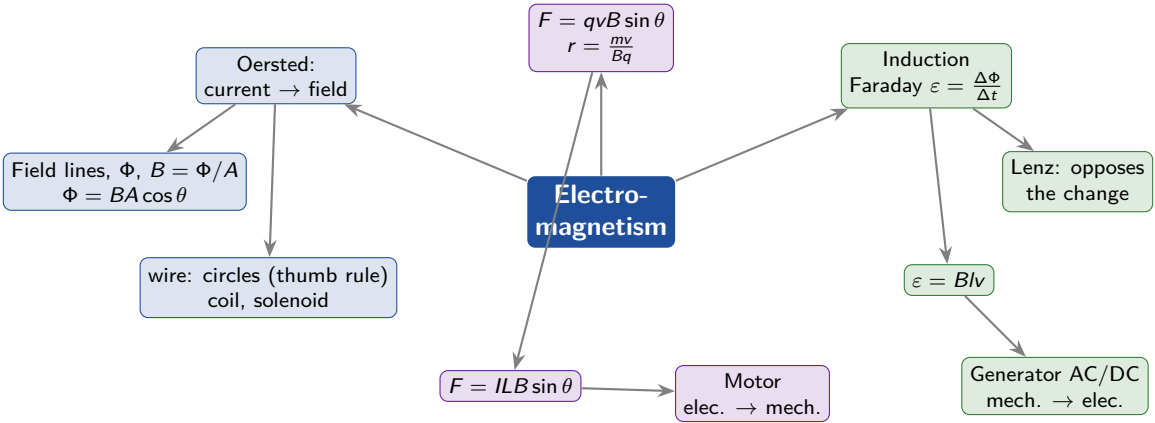
Feature	Electric motor	Generator (dynamo)
Energy change	electrical $\rightarrow$ mechanical	mechanical $\rightarrow$ electrical
Principle	force on a current-carrying coil in a field ($F = ILB$)	electromagnetic induction ($\varepsilon = \Delta\Phi/\Delta t$)
Input	current from a battery/supply	rotation of the coil
Output	rotation	induced current
Rings	split rings: reverse current every half turn	AC: full slip rings; DC: split rings
Hand rule	right hand rule / Fleming's left hand rule	Lenz's law gives the current direction

All derivations of this chapter at a glance

Revise

No.	Result	Key idea	Uses
1	$\Phi = BA\cos\theta$	effective area $\perp$ field is $A\cos\theta$	$B = \Phi/A$
2	$r = \frac{mv}{Bq}$, $T = \frac{2\pi m}{Bq}$	magnetic force = centripetal force	$F = qvB$
3	$F = ILB\sin\theta$	add qvB over all charges, $v = L/t$, $I = Q/t$	$F = qvB$
4	$\varepsilon = \frac{\Delta\Phi}{\Delta t}$, $\varepsilon = Blv$	work done = electrical energy; $\Delta\Phi = Bls$; $P = \varepsilon I = Fv$	$F = BIl$

Concept map


Electro-magnetism
Oersted: current → field
Field lines, Φ, $B = \Phi/A$ $\Phi = BA\cos\theta$
wire: circles (thumb rule) coil, solenoid
$F = qvB\sin\theta$ $r = \frac{mv}{Bq}$
$F = ILB\sin\theta$
Motor elec. → mech.
Induction Faraday $\varepsilon = \frac{\Delta\Phi}{\Delta t}$
Lenz: opposes the change
$\varepsilon = Blv$
Generator AC/DC mech. → elec.

Formula sheet

Must memorise

Quantity	Formula / unit
Magnetic flux density	$B = \dfrac{\Phi}{A}$; tesla (T) $=$ Wb/m^2 $=$ N/(A m)
Magnetic flux	$\Phi = BA\cos\theta$; weber (Wb)
Force on a moving charge	$F = qvB\sin\theta$ $(= qvB$ when $v \perp B)$
Charge in a uniform field	$r = \dfrac{mv}{Bq}$; $T = \dfrac{2\pi m}{Bq}$
Force on a current	$F = ILB\sin\theta$ $(= ILB$ when $I \perp B)$
Faraday's law	$\varepsilon = \dfrac{\Delta\Phi}{\Delta t}$; N turns: $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t}$
Motional emf	$\varepsilon = Blv$

What we have learnt (textbook summary)

- **Magnetic flux**: a measure of the number of field lines through an area.
- **Magnetic flux density** B: flux through a plane perpendicular to the field $\div$ area of the plane.
- A current-carrying wire produces a magnetic field.
- $F = qvB\sin\theta$ and $F = ILB\sin\theta$.
- In an electric **motor**, electrical energy is converted into mechanical energy.
- Producing current by relative motion between a coil and a magnetic field is **electromagnetic induction**.
- **Faraday's law**: induced emf in a closed loop = rate of change of magnetic flux through it.
- **Lenz's law**: the induced current opposes the change in flux.
- A conductor of length l moving $\perp$ to B with speed v develops the **motional emf** Blv.
- In **generators**, mechanical energy is converted into electrical energy.

1-mark questions

1M

1. Who showed a current affects a compass? **Oersted**
2. SI unit of magnetic flux? **weber**
3. SI unit of flux density? **tesla (Wb/m^2)**
4. Shape of field lines of a straight wire? **concentric circles**
5. Flux through a plane parallel to B? **zero**
6. Force on a charge moving along B? **zero**
7. Device like a bar magnet made of coils? **solenoid**
8. Energy change in a motor? **electrical $\rightarrow$ mechanical**
9. Part that reverses current in a motor? **split rings**
10. Law giving the direction of induced current? **Lenz's law**
11. Lenz's law follows from? **conservation of energy**
12. Formula for motional emf? $\varepsilon = Blv$
13. Stove based on induction? **induction stove**
14. Rings used in an AC generator? **full slip rings**

Textbook MCQs

1M

1. Converts electrical energy into mechanical energy:
   a) motor b) battery c) generator d) switch
   Answer: a

2. Converts mechanical energy into electrical energy:
   a) motor b) battery c) generator d) switch
   Answer: c

3. Force on a wire placed perpendicular to a uniform field:
   a) 0 b) ILB c) $2ILB$ d) $ILB/2$
   Answer: b ($\sin 90^\circ = 1$)

4. One tesla =
   a) N/C b) N/(A m) c) A/m d) N/(A s)
   Answer: b ($B = F/IL$)

5. Unit of magnetic flux:
   a) dyne b) oersted c) gauss d) weber
   Answer: d

6. No force acts on a current-carrying conductor kept
   a) parallel to the field b) perpendicular to the field c) in the field d) away from the field
   Answer: a

Reflections on concepts: figure questions

1M 2M

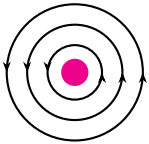


fig-Q1

Q2. Field lines as in fig-Q1 (anti-clockwise). Direction of current?
Thumb rule: anticlockwise field ⇒ current **out of the page** (towards the reader).

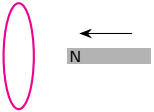


fig-Q2

Q3. N pole moves towards the coil. What happens to the flux?
Flux through the coil **increases**; a current is induced so that the near face becomes N (Lenz).

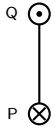


fig-Q3

Q4. Coil ⊥ page: current into page at P, out at Q. Direction of the field of the coil?
Between P and Q both wires give a field to the **right**: the field is along the axis, from left to right (in the plane of the page, ⊥ to the coil).

Q1. Are field lines closed? Yes. For a straight wire they are circles; for a solenoid/bar magnet, lines outside (N → S) continue inside (S → N), forming closed loops.

Application of concepts: model answers

2M



fig-Q4

1. Current as in fig-Q4 (anticlockwise, as seen). Pole at this face?
North pole: anticlockwise current ⇒ field points towards the viewer.

2. Why is a TV picture distorted near a magnet?
The picture is made by a moving electron beam; the magnet's field exerts a force $F = qvB\sin\theta$ on the electrons and deflects them from their correct spots.

5. Coil and magnet move in the **same direction** with the same speed (fig-Q6). What happens?
No relative motion ⇒ flux through the coil does not change ⇒ **no induced current**.

6. Applications of Faraday's law:
generators, security metal detectors, tape recorder heads, induction stoves, ATM card readers, transformers.

Reflections Q2 of electromagnet ideas. Why use a solenoid for an electromagnet?
Its field inside is strong and uniform like a bar magnet's and can be switched on/off with the current.

Lenz and energy. Why is it harder to push a magnet into a closed coil than into an open one?
In a closed coil an induced current flows and repels the magnet; our extra work becomes electrical energy. In an open coil no current flows, so no opposing force.

4-mark questions: answer points

4M

Q1. Two activities: a current-carrying wire produces a field.

Activity 1 (Oersted: compass under wire deflects when switched on) · Activity 4 (compasses around a vertical wire set as tangents; iron filings form circles) · diagrams · conclusion.

Q2. Verify the force on a wire in a field.

Activity 8: wire between split sticks, 9 V battery, horseshoe magnet · wire deflects · reverse poles/current $\Rightarrow$ deflection reverses · right hand rule · $F = ILB$.

Q3. Explain Faraday's law with an activity.

Activity 9: coil + galvanometer, magnet in/out, at rest, S pole · law $\varepsilon = \Delta\Phi/\Delta t$ · faster motion, bigger deflection · N turns.

Q4. Experiment for Faraday's law: instruments, tips, precautions.

Coil of many turns, strong bar magnet, sensitive galvanometer, wires · **tips**: more turns, stronger magnet, faster motion · **precautions**: tight connections, galvanometer away from the magnet, note zero first, move magnet along the axis.

Q5. Working of an electric motor.

fig-12 · forces on AB, CD equal and opposite $\Rightarrow$ couple · split rings and brushes reverse current each half turn · continuous rotation · electrical $\rightarrow$ mechanical.

Q6. Derive $\varepsilon = Blv$ / explain AC generator.

Derivation 4 (eqs 8–14) **or** fig-17: rotating coil, changing flux, slip rings, emf zero/max, AC graph.

Practice numericals (homework / class test)

2M

1. A flux of 0.06 Wb passes normally through 0.3 m^2. Find B. — 0.2 T
2. $B = 0.4$ T, area 0.5 m^2, normal at 60° to B. Find Φ. — 0.1 Wb
3. A 2 m wire carries 3 A perpendicular to $B = 0.5$ T. Find F. — 3 N
4. An electron (1.6×10^{-19} C) moves at 5×10^6 m/s $\perp$ $B = 0.2$ T. Find F. — 1.6×10^{-13} N
5. A 0.5 m rod moves at 4 m/s $\perp$ $B = 0.25$ T. Find the emf. — 0.5 V
6. Flux per turn in a 200-turn coil changes by 5×10^{-3} Wb in 0.2 s. Find ε. — 5 V
7. Show that the time period of a charge in a uniform field is independent of its speed. — $T = 2\pi m/Bq$

Practice numericals: worked solutions

2M

1. $B = \dfrac{\Phi}{A} = \dfrac{0.06}{0.3} = \boxed{0.2\ \text{T}}$
2. $\Phi = BA\cos 60^\circ = 0.4 \times 0.5 \times \frac{1}{2} = \boxed{0.1\ \text{Wb}}$
3. $F = ILB = 3 \times 2 \times 0.5 = \boxed{3\ \text{N}}$
4. $F = qvB = 1.6 \times 10^{-19} \times 5 \times 10^{6} \times 0.2$
   $= \boxed{1.6 \times 10^{-13}\ \text{N}}$
5. $\varepsilon = Blv = 0.25 \times 0.5 \times 4 = \boxed{0.5\ \text{V}}$
6. $\varepsilon = N\dfrac{\Delta\Phi_0}{\Delta t} = \dfrac{200 \times 5 \times 10^{-3}}{0.2} = \dfrac{1}{0.2} = \boxed{5\ \text{V}}$
7. $qvB = \dfrac{mv^2}{r} \Rightarrow r = \dfrac{mv}{Bq}$; $T = \dfrac{2\pi r}{v} = \dfrac{2\pi m}{Bq}$: no v in the result (Derivation 2).

Important diagrams and common mistakes

4M

Must practise

1. Field lines of a bar magnet – fig-2
2. Field of a straight wire ($\odot$ and $\otimes$) – fig-4(b),(c)
3. Field of a solenoid with N, S – fig-6(b)
4. Right hand rule for force – fig-8(b)
5. Coil in a field with forces – fig-12(a); motor – fig-12(c)
6. Faraday's activity – fig-13; cross-wire set-up – fig-15
7. AC generator and emf graph – fig-17(a),(b)

★ Common mistakes that lose marks

- No arrows on field lines.
- Using $\sin\theta$ with θ between B and the **plane** instead of the **normal** in $\Phi = BA\cos\theta$.
- Forgetting $F = 0$ when the wire/charge is parallel to B.
- Not reversing the force for an electron.
- Using cm or cm^2 in formulas (convert to m, m^2).
- Forgetting N in $\varepsilon = N\,\Delta\Phi_0/\Delta t$.
- Mixing up motor (split rings) and AC generator (slip rings).

Suggested experiments and projects

Suggested experiments

1. Two activities showing that a current-carrying wire produces a magnetic field.
2. Verify that a current-carrying conductor experiences a force in a magnetic field.
3. Explain Faraday's law of induction with an activity.
4. Suggest an experiment for Faraday's law: instruments, tips and precautions.
5. Verify with an experiment that a current-carrying wire produces a field.

Suggested projects

1. Collect information on generating current using Faraday's law.
2. Materials and procedure for making a simple electric motor; write a report.
3. Collect information on Faraday's experiments.

Thank you!

Keywords to revise:

magnetic flux · magnetic flux density · field lines · solenoid · electric motor · slip rings
induced current · induced emf · Faraday's law · Lenz's law · motional emf · AC and DC generators